

circuit board as a three-dimensional (3D) model. This would help you collaborate with mechanical designers. With support of rigid-flex design capabilities in the latest EDA tools, you can visualise the design in 3D and help check for mechanical issues like component collisions and clearance issues.

Open libraries are kind of a sour area. Some EDA tools also offer integrated open source libraries to help pick available components to start with the design. Some online tools provide availability and pricing across vendors. However, Hrishikesh Kamat, chief executive officer, Shalaka Technologies, feels that, to get the best in design, you need to manually build the components instead of blindly trusting open source libraries. He adds, "We have our own library to go to in case of non-regular components."

Anuj Deshpande, founder,

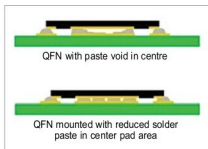


Fig. 2: Improper application of solder resulting in gaps

Makerville Solutions, also believes in designing the components manually. He says, "A lot of times this can increase the timeline of the project, but the output is very reliable." However, the saving grace for open libraries, as Deshpande explains, is the extensive use of standard components that are easily available in open libraries.

Kamat explains some designing issues his team faces, "We have to take care while defining the power

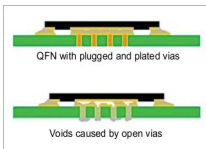


Fig. 3: Open vias need to be taken care of

and ground planes, and avoid auto-correct most of the times." Auto-correct, on one hand, helps in getting the design right but, on the other, can cause unexpected problems while manufacturing.

Problems in manufacturing

When talking about manufacturing printed boards, the magic happens due to very minute particles. Dana P. Ilmer, Superior Flux & Manufacturing Co., says, "You have to be

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